Laporan Pemrograman Mobile 6



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Pertanyaan Refleksi

1. Apa perbedaan animasi implisit dan eksplisit? Kapan memilih masing-masing?
2. Mengapa dua Hero harus memiliki tag yang sama, dan apa yang terjadi bila tag-nya kembali muncul
dua kali di halaman yang sama?
3. Apa keuntungan memakai Theme.of(context) dibanding menulis warna langsung pada widget?
4. Apa perbedaan berpindah tab dengan IndexedStack dan membuka halaman dengan Navigator?
5. Mengapa AnimationController wajib di-dispose?

Jawab :

1. Animasi Implisit adalah animasi yang dikelola oleh flutter dengan mengubah setState() lalu flutter akan mengurus transisinya, biasanya dipilih untuk perubahan warna, ukuran, transparansi. Animasi Eksplisit adalah animasi yang membutuhkan kontrol penuh dan biasa dipilih untuk detail animasi yang looping.

2. Hero menggunakan tag sebagai pengenal unik agar flutter tahu elemen di halaman yang bertransisi. Jika tag muncul 2 kali di halaman yang sama flutter akan mengalami assertion failure karena tagnya jadi tidak unik lagi, flutter jadi tidak bisa menentukan elemen yang menjadi awal transisi.

3. Theme.of(context) mendukung mode gelap dan terang sehingga aplikasi dapat otomatis menyesuaikan warna, desainnya konsisten juga saat diubah ke mode gelap dan terang, dan pemeliharaannya lebih mudah dibanding mengganti warna per widget.

4. IndextedStack menjaga state seluruh halaman tetap ada di memori saat berpindah tab sedangkan Navigator menumpuk halaman baru diatas halaman lama atau menghapusnya.

5. AnimationController menggunakan ticker yang terus berjalan dan memakan memori sehingga jika tidak di dispose makan akan menyebabkan memory leak karna controller mencoba memperbarui animasi di latar belakang walau sudah tidak ada.

Bagian A :

```dart
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});
```

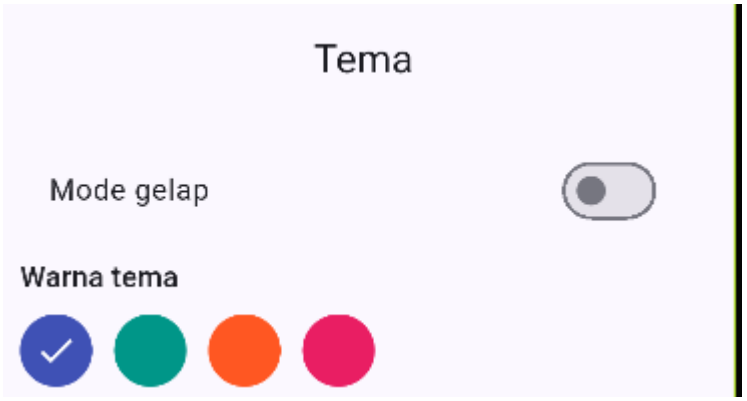
```dart
 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         home: Scaffold(
           appBar: AppBar(title: const Text('Tema')),
           body: const PengaturanTab(),
         ),
       );
     },
   );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

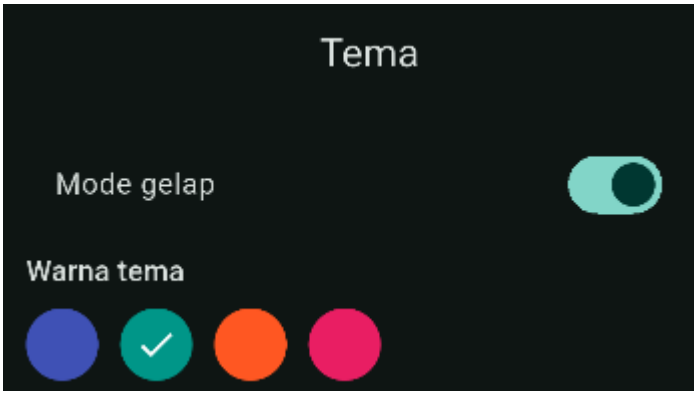
 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
             },
           ),
           const SizedBox(height: 8),
           Text(
             'Warna tema',
             style: Theme.of(context).textTheme.titleMedium,
           ),
           const SizedBox(height: 8),
           Wrap(
             spacing: 12,
             children: [
               for (final c in pilihanWarna)
                 GestureDetector(
                   onTap: () => seedColor.value = c,
                   child: CircleAvatar(
```

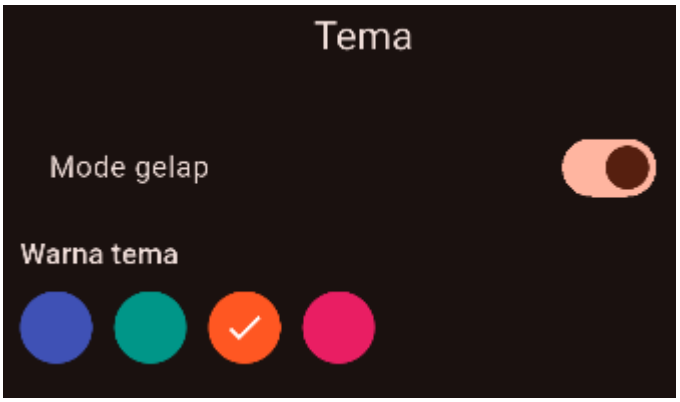
```
                    backgroundColor: c,
                    child: seedColor.value == c
                        ? const Icon(Icons.check, color: Colors.white)
                        : null,
                  ),
                ),
              ],
            ),
          ],
        );
      },
    );
  }
}
```

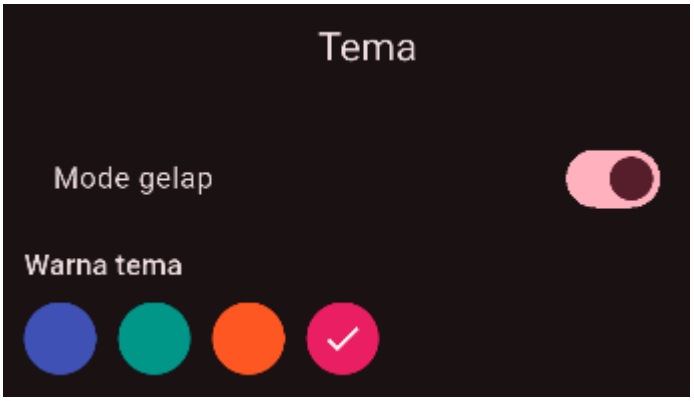
Hasil :




Tema
Mode gelap
Warna tema


Tema
Mode gelap
Warna tema


Tema
Mode gelap
Warna tema


Tema
Mode gelap
Warna tema

Bagian B :

```dart
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

class Item {
```

```dart
 final String nama;
 final IconData ikon;
 final Color warna;

 const Item(this.nama, this.ikon, this.warna);
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (context) => Scaffold(
             appBar: AppBar(title: const Text('Beranda')),
             body: const BerandaTab(),
           ),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final item = settings.arguments as Item;
             return MaterialPageRoute(
               builder: (context) => DetailPage(item: item),
               settings: settings,
             );
           }
           return null;
         },
       );
     },
   );
 }
}
```

```dart
class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListView.builder(
     itemCount: daftarItem.length,
     itemBuilder: (context, i) {
       final item = daftarItem[i];
       return ListTile(
         leading: Hero(
           tag: 'ikon-${item.nama}',
           child: CircleAvatar(
             backgroundColor: item.warna,
             child: Icon(item.ikon, color: Colors.white),
           ),
         ),
         title: Text(item.nama),
         trailing: const Icon(Icons.chevron_right),
         onTap: () {
           Navigator.pushNamed(context, '/detail', arguments: item);
         },
       );
     },
   );
 }
}

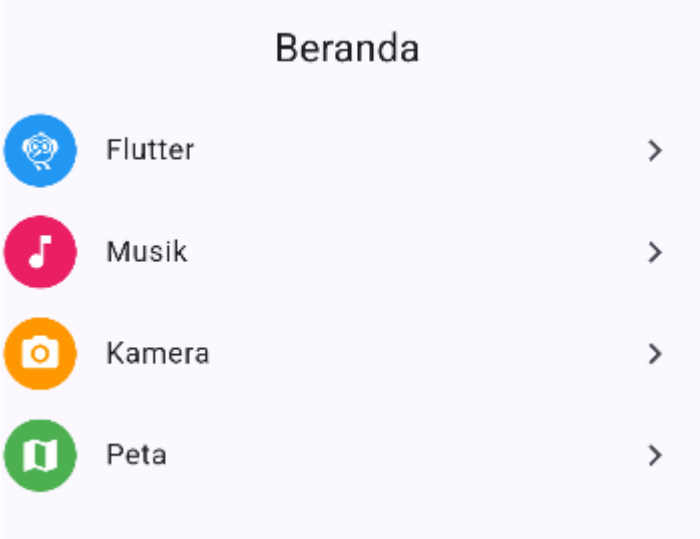
class DetailPage extends StatelessWidget {
 final Item item;

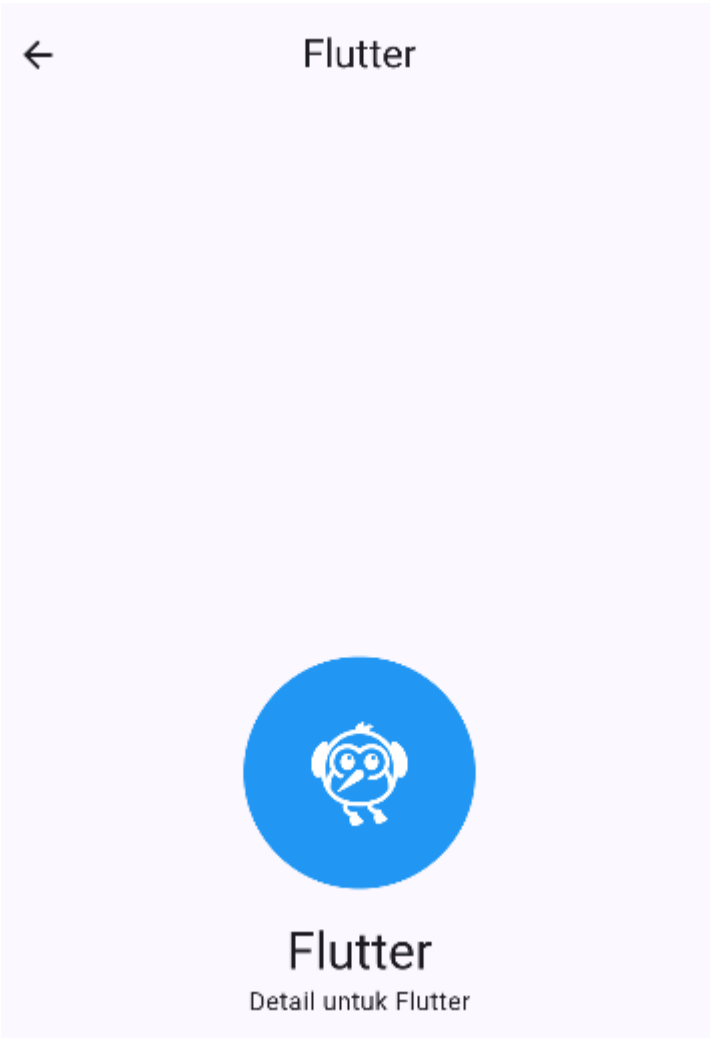
 const DetailPage({super.key, required this.item});

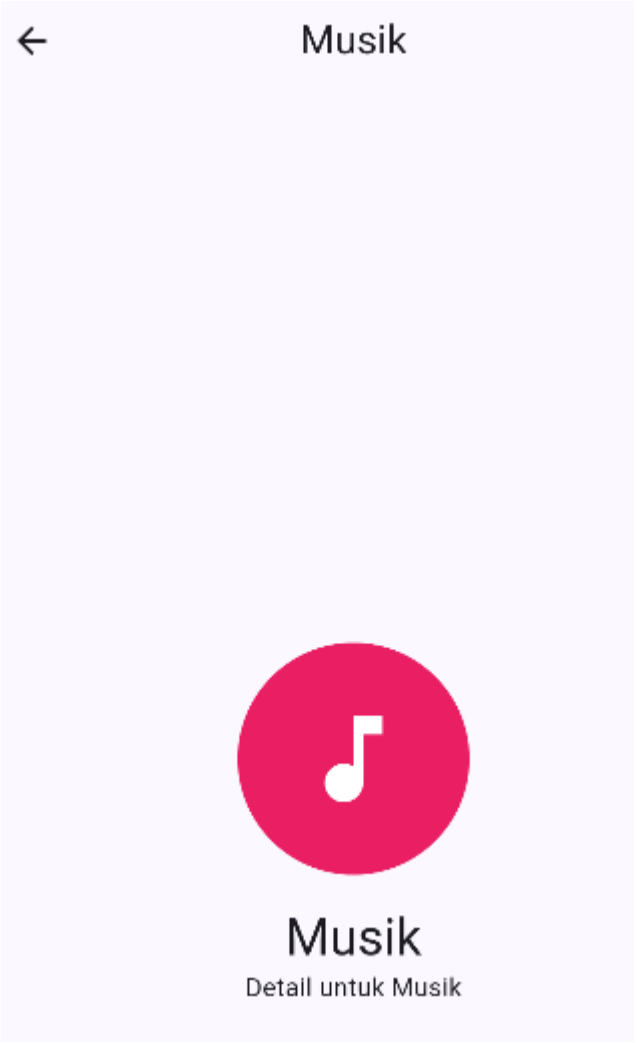
 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   return Scaffold(
     appBar: AppBar(title: Text(item.nama)),
     body: Center(
       child: Column(
         mainAxisSize: MainAxisSize.min,
         children: [
           Hero(
             tag: 'ikon-${item.nama}',
             child: CircleAvatar(
               radius: 64,
               backgroundColor: item.warna,
               child: Icon(item.ikon, size: 64, color: Colors.white),
             ),
           ),
           const SizedBox(height: 16),
           Text(item.nama, style: tema.textTheme.headlineMedium),
```

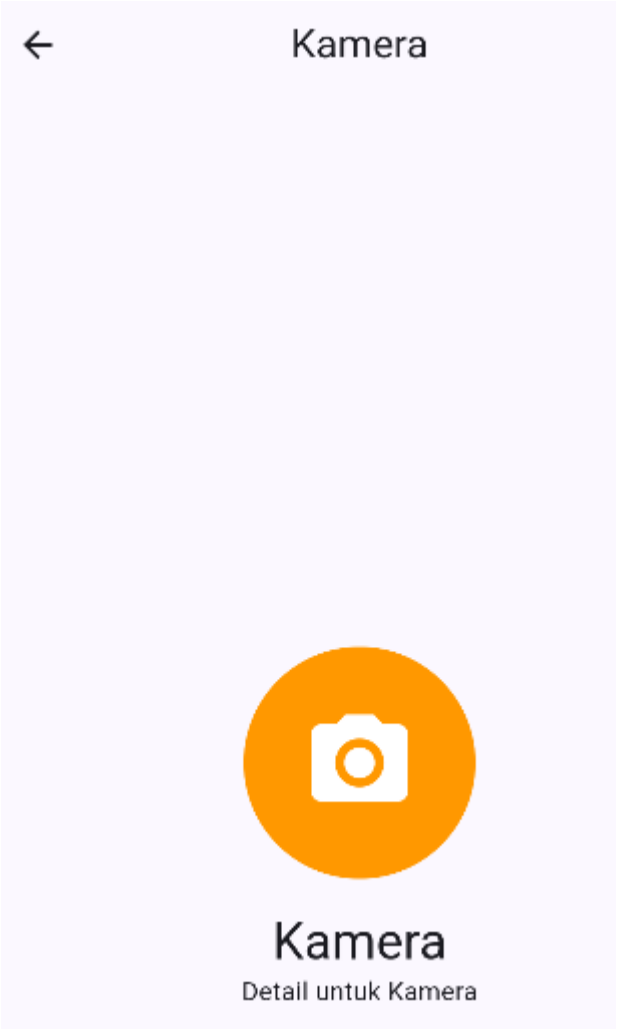
```
            Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
          ],
        ),
      ),
    );
  }
}
```

Hasil :




Flutter
Flutter
Detail untuk Flutter


Musik
Musik
Detail untuk Musik


Kamera
Kamera
Detail untuk Kamera

Bagian C :

```dart
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         home: Scaffold(
           appBar: AppBar(title: const Text('Animasi')),
           body: const AnimasiTab(),
         ),
       );
     },
   );
 }
}

class AnimasiTab extends StatefulWidget {
 const AnimasiTab({super.key});

 @override
 State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
   with SingleTickerProviderStateMixin {
 bool _besar = false;
 int _hitung = 0;
 late final AnimationController _putar;
```

```dart
@override
void initState() {
  super.initState();
  _putar = AnimationController(
    vsync: this,
    duration: const Duration(seconds: 2),
  )..repeat();
}

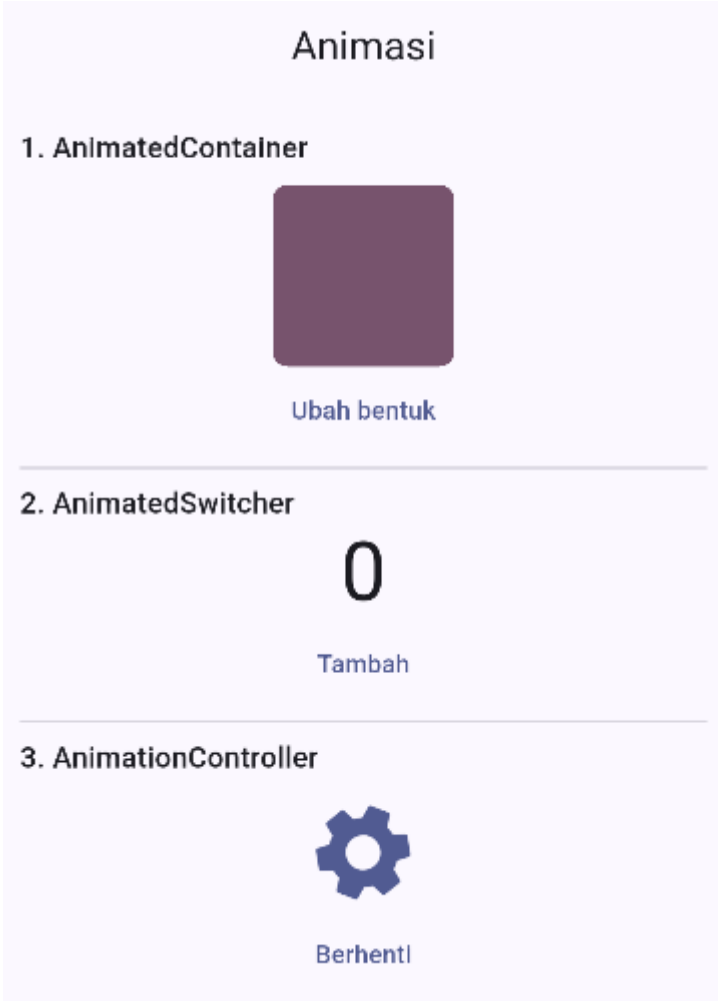
@override
void dispose() {
  _putar.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  final tema = Theme.of(context);
  final warna = tema.colorScheme;

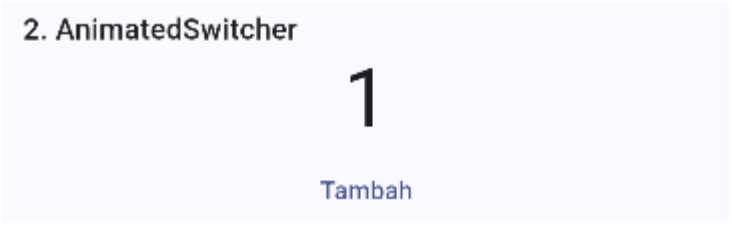
  return ListView(
    padding: const EdgeInsets.all(16),
    children: [
      Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedContainer(
          duration: const Duration(milliseconds: 500),
          curve: Curves.easeInOut,
          width: _besar ? 200 : 100,
          height: 100,
          decoration: BoxDecoration(
            color: _besar ? warna.primary : warna.tertiary,
            borderRadius: BorderRadius.circular(_besar ? 50 : 8),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _besar = !_besar),
        child: const Text('Ubah bentuk'),
      ),
      const Divider(),
      Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
      Center(
        child: AnimatedSwitcher(
          duration: const Duration(milliseconds: 300),
          transitionBuilder: (child, anim) {
            return ScaleTransition(scale: anim, child: child);
          },
          child: Text(
            '$_hitung',
            key: ValueKey(_hitung),
```

```
              style: tema.textTheme.displayMedium,
            ),
          ),
        ),
        TextButton(
          onPressed: () => setState(() => _hitung++),
          child: const Text('Tambah'),
        ),
        const Divider(),
        Text('3. AnimationController', style: tema.textTheme.titleMedium),
        const SizedBox(height: 8),
        Center(
          child: RotationTransition(
            turns: _putar,
            child: Icon(Icons.settings, size: 64, color: warna.primary),
          ),
        ),
        TextButton(
          onPressed: () {
            if (_putar.isAnimating) {
              _putar.stop();
            } else {
              _putar.repeat();
            }
            setState(() {});
          },
          child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
        ),
      ],
    );
  }
}
```

Hasil :


Animasi
1. AnimatedContainer
Ubah bentuk
2. AnimatedSwitcher
0
Tambah
3. AnimationController
Berhenti


1. AnimatedContainer
Ubah bentuk


2. AnimatedSwitcher
1
Tambah


3. AnimationController
Putar

Bagian D :

```dart
import 'package:flutter/material.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

class Item {
 final String nama;
 final IconData ikon;
 final Color warna;

 const Item(this.nama, this.ikon, this.warna);
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
```

```
        initialRoute: '/',
        routes: {
          '/': (context) => const ShellPage(),
        },
        onGenerateRoute: (settings) {
          if (settings.name == '/detail') {
            final item = settings.arguments as Item;
            return MaterialPageRoute(
              builder: (context) => DetailPage(item: item),
              settings: settings,
            );
          }
          return null;
        },
      );
    },
  );
 }
}

class ShellPage extends StatefulWidget {
 const ShellPage({super.key});

 @override
 State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
 int _index = 0;

 static const _halaman = [
   BerandaTab(),
   AnimasiTab(),
   PengaturanTab(),
 ];

 static const _judul = ['Beranda', 'Animasi', 'Pengaturan'];

 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(title: Text(_judul[_index])),
     body: IndexedStack(index: _index, children: _halaman),
     bottomNavigationBar: NavigationBar(
       selectedIndex: _index,
       onDestinationSelected: (i) => setState(() => _index = i),
       destinations: const [
         NavigationDestination(
           icon: Icon(Icons.home_outlined),
           selectedIcon: Icon(Icons.home),
           label: 'Beranda',
         ),
         NavigationDestination(
```

```
          icon: Icon(Icons.animation),
          label: 'Animasi',
        ),
        NavigationDestination(
          icon: Icon(Icons.settings_outlined),
          selectedIcon: Icon(Icons.settings),
          label: 'Pengaturan',
        ),
      ],
    ),
  );
 }
}

class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListView.builder(
     itemCount: daftarItem.length,
     itemBuilder: (context, i) {
       final item = daftarItem[i];
       return ListTile(
         leading: Hero(
           tag: 'ikon-${item.nama}',
           child: CircleAvatar(
             backgroundColor: item.warna,
             child: Icon(item.ikon, color: Colors.white),
           ),
         ),
         title: Text(item.nama),
         trailing: const Icon(Icons.chevron_right),
         onTap: () {
           Navigator.pushNamed(context, '/detail', arguments: item);
         },
       );
     },
   );
 }
}

class DetailPage extends StatelessWidget {
 final Item item;

 const DetailPage({super.key, required this.item});

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   return Scaffold(
     appBar: AppBar(title: Text(item.nama)),
     body: Center(
```

```
        child: Column(
          mainAxisSize: MainAxisSize.min,
          children: [
            Hero(
              tag: 'ikon-${item.nama}',
              child: CircleAvatar(
                radius: 64,
                backgroundColor: item.warna,
                child: Icon(item.ikon, size: 64, color: Colors.white),
              ),
            ),
            const SizedBox(height: 16),
            Text(item.nama, style: tema.textTheme.headlineMedium),
            Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
          ],
        ),
      ),
    );
  }
}

class AnimasiTab extends StatefulWidget {
 const AnimasiTab({super.key});

 @override
 State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
    with SingleTickerProviderStateMixin {
 bool _besar = false;
 int _hitung = 0;
 late final AnimationController _putar;

 @override
 void initState() {
   super.initState();
   _putar = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 2),
   )..repeat();
 }

 @override
 void dispose() {
   _putar.dispose();
   super.dispose();
 }

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
```

```dart
  final warna = tema.colorScheme;

  return ListView(
    padding: const EdgeInsets.all(16),
    children: [
      Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedContainer(
          duration: const Duration(milliseconds: 500),
          curve: Curves.easeInOut,
          width: _besar ? 200 : 100,
          height: 100,
          decoration: BoxDecoration(
            color: _besar ? warna.primary : warna.tertiary,
            borderRadius: BorderRadius.circular(_besar ? 50 : 8),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _besar = !_besar),
        child: const Text('Ubah bentuk'),
      ),
      const Divider(),
      Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
      Center(
        child: AnimatedSwitcher(
          duration: const Duration(milliseconds: 300),
          transitionBuilder: (child, anim) {
            return ScaleTransition(scale: anim, child: child);
          },
          child: Text(
            '$_hitung',
            key: ValueKey(_hitung),
            style: tema.textTheme.displayMedium,
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _hitung++),
        child: const Text('Tambah'),
      ),
      const Divider(),
      Text('3. AnimationController', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: RotationTransition(
          turns: _putar,
          child: Icon(Icons.settings, size: 64, color: warna.primary),
        ),
      ),
      TextButton(
        onPressed: () {
```

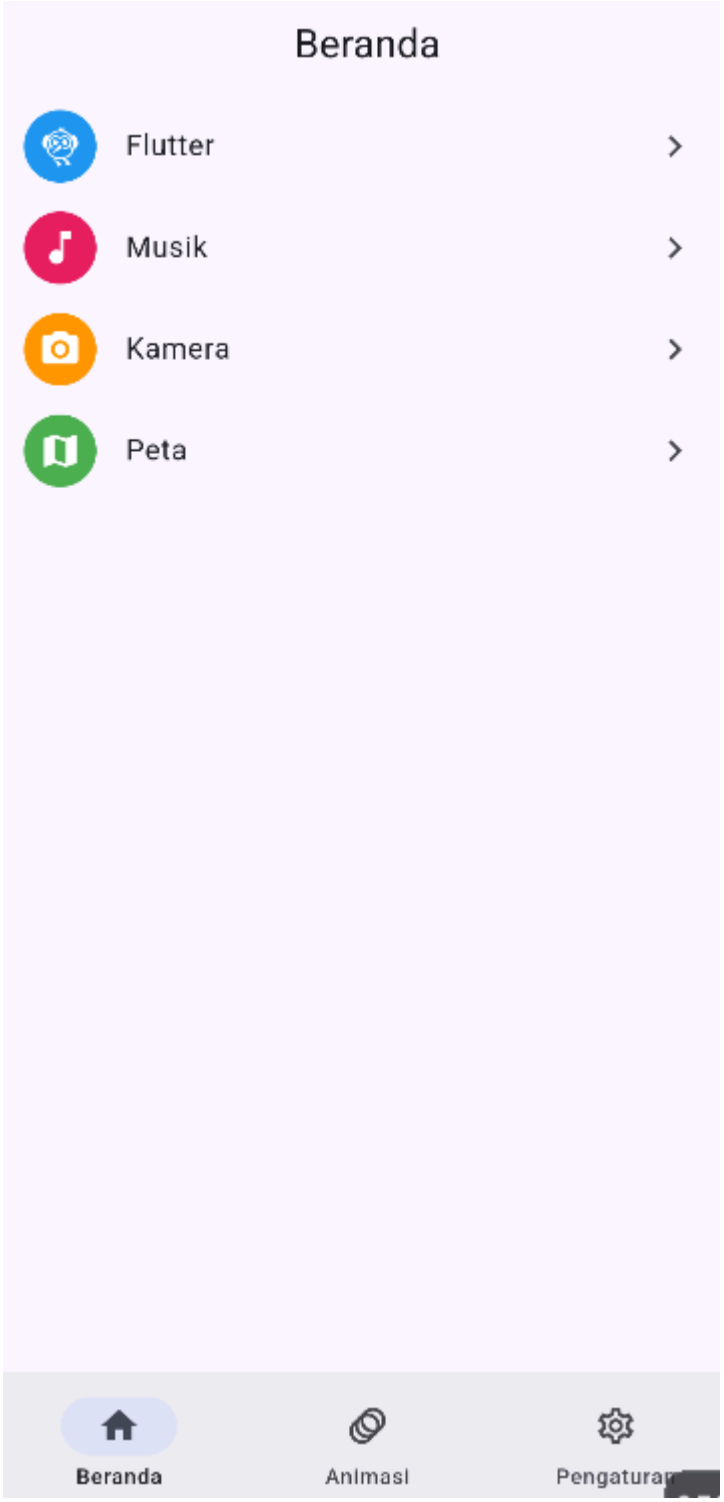
```
          if (_putar.isAnimating) {
            _putar.stop();
          } else {
            _putar.repeat();
          }
          setState(() {});
        },
        child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
      ),
    ],
  );
 }
}

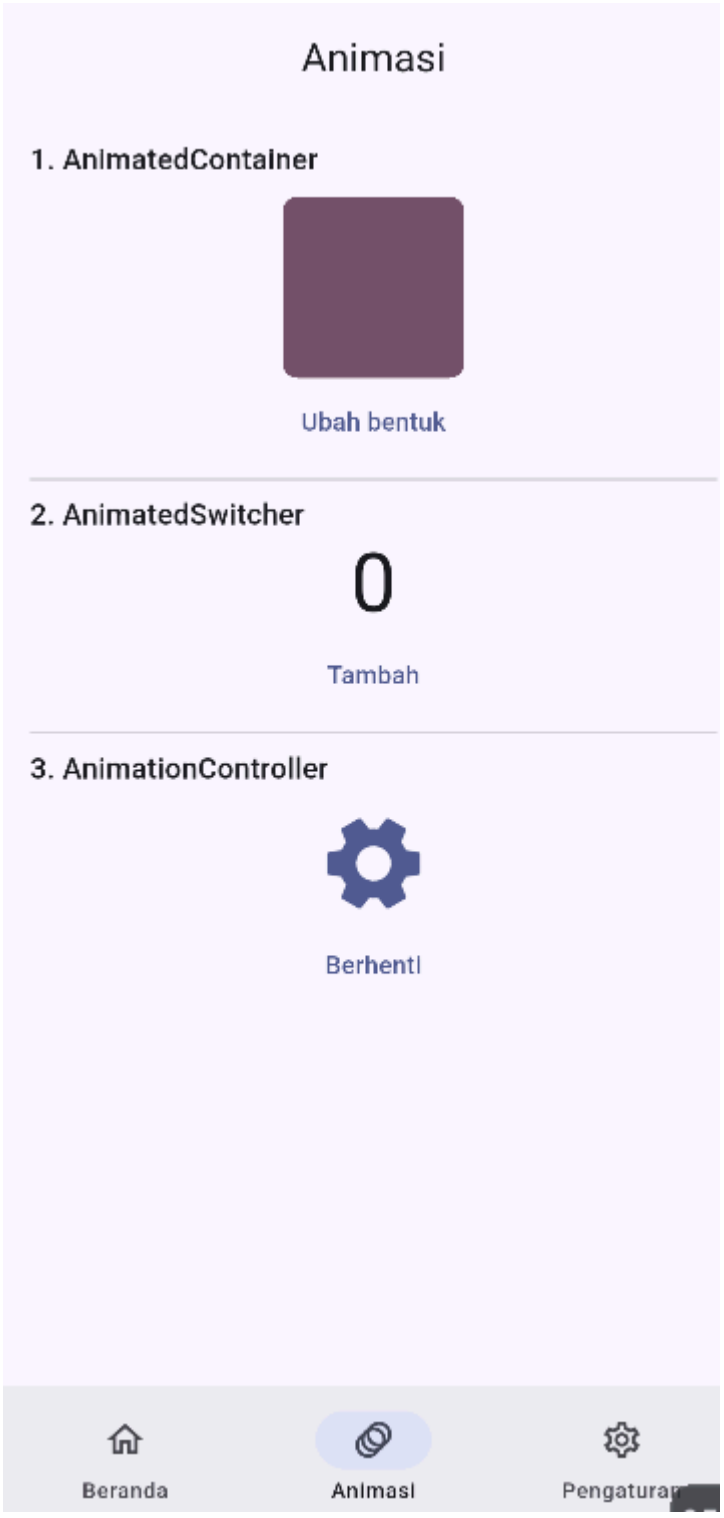
class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
             },
           ),
           const SizedBox(height: 8),
           Text(
             'Warna tema',
             style: Theme.of(context).textTheme.titleMedium,
           ),
           const SizedBox(height: 8),
           Wrap(
             spacing: 12,
             children: [
               for (final c in pilihanWarna)
                 GestureDetector(
                   onTap: () => seedColor.value = c,
                   child: CircleAvatar(
                     backgroundColor: c,
                     child: seedColor.value == c
                         ? const Icon(Icons.check, color: Colors.white)
                         : null,
                   ),
                 ),
             ],
           ),
```

```
          ],
        );
      },
    );
  }
}
```

Hasil :


Beranda
Flutter
Musik
Kamera
Peta
Beranda
Animasi
Pengaturan


Animasi
1. AnimatedContainer
Ubah bentuk
2. AnimatedSwitcher
0
Tambah
3. AnimationController
Berhenti
Beranda
Animasi
Pengaturan


Pengaturan
Mode gelap
Warna tema
Beranda
Animasi
Pengaturan

Latihan Mandiri :

```dart
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.indigo);

const pilihanWarna = [
 Colors.indigo,
 Colors.teal,
 Colors.deepOrange,
 Colors.pink,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

class Item {
 final String nama;
 final IconData ikon;
 final Color warna;

 const Item(this.nama, this.ikon, this.warna);
}

const daftarItem = [
 Item('Flutter', Icons.flutter_dash, Colors.blue),
 Item('Musik', Icons.music_note, Colors.pink),
 Item('Kamera', Icons.camera_alt, Colors.orange),
 Item('Peta', Icons.map, Colors.green),
];

// Latihan 1: Memuat tema tersimpan saat aplikasi dimulai
void main() async {
 WidgetsFlutterBinding.ensureInitialized();
 final prefs = await SharedPreferences.getInstance();

 final isDark = prefs.getBool('isDark') ?? false;
 themeMode.value = isDark ? ThemeMode.dark : ThemeMode.light;

 final colorValue = prefs.getInt('seedColor');
 if (colorValue != null) {
   seedColor.value = Color(colorValue);
 }

 runApp(const MyApp());
}
```

```dart
class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Praktikum 6 - Latihan Mandiri',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (context) => const ShellPage(),
         },
         // Latihan 3: Transisi Halaman dengan FadeTransition
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final item = settings.arguments as Item;
             return PageRouteBuilder(
               settings: settings,
               pageBuilder: (context, animation, secondaryAnimation) =>
                   DetailPage(item: item),
               transitionsBuilder:
                   (context, animation, secondaryAnimation, child) {
                 return FadeTransition(
                   opacity: animation,
                   child: child,
                 );
               },
             );
           }
           return null;
         },
         // Latihan 4: Halaman 404 (onUnknownRoute)
         onUnknownRoute: (settings) {
           return MaterialPageRoute(
             builder: (context) => Scaffold(
               appBar: AppBar(title: const Text('404')),
               body: const Center(
                 child: Text(
                   '404 - Halaman tidak ditemukan',
                   style: TextStyle(fontSize: 18),
                 ),
               ),
             ),
           );
         },
       );
```

```
    },
  );
 }
}

class ShellPage extends StatefulWidget {
 const ShellPage({super.key});

 @override
 State<ShellPage> createState() => _ShellPageState();
}

class _ShellPageState extends State<ShellPage> {
 int _index = 0;

 static const _halaman = [
   BerandaTab(),
   AnimasiTab(),
   PengaturanTab(),
 ];

 static const _judul = ['Beranda', 'Animasi', 'Pengaturan'];

 void _pindahTab(int i) {
   setState(() => _index = i);
 }

 @override
 Widget build(BuildContext context) {
   return Scaffold(
     appBar: AppBar(title: Text(_judul[_index])),
     // Latihan 2: Drawer navigasi
     drawer: Drawer(
       child: ListView(
         padding: EdgeInsets.zero,
         children: [
           DrawerHeader(
             decoration: BoxDecoration(
               color: Theme.of(context).colorScheme.primaryContainer,
             ),
             child: Text(
               'Menu Navigasi',
               style: Theme.of(context).textTheme.headlineSmall,
             ),
           ),
           ListTile(
             leading: const Icon(Icons.home),
             title: const Text('Beranda'),
             selected: _index == 0,
             onTap: () {
               _pindahTab(0);
               Navigator.pop(context);
             },
```

```
          ),
          ListTile(
            leading: const Icon(Icons.animation),
            title: const Text('Animasi'),
            selected: _index == 1,
            onTap: () {
              _pindahTab(1);
              Navigator.pop(context);
            },
          ),
          ListTile(
            leading: const Icon(Icons.settings),
            title: const Text('Pengaturan'),
            selected: _index == 2,
            onTap: () {
              _pindahTab(2);
              Navigator.pop(context);
            },
          ),
        ],
      ),
    ),
    body: IndexedStack(index: _index, children: _halaman),
    bottomNavigationBar: NavigationBar(
      selectedIndex: _index,
      onDestinationSelected: _pindahTab,
      destinations: const [
        NavigationDestination(
          icon: Icon(Icons.home_outlined),
          selectedIcon: Icon(Icons.home),
          label: 'Beranda',
        ),
        NavigationDestination(
          icon: Icon(Icons.animation),
          label: 'Animasi',
        ),
        NavigationDestination(
          icon: Icon(Icons.settings_outlined),
          selectedIcon: Icon(Icons.settings),
          label: 'Pengaturan',
        ),
      ],
    ),
  );
 }
}

class BerandaTab extends StatelessWidget {
 const BerandaTab({super.key});

 @override
 Widget build(BuildContext context) {
   return ListView.builder(
```

```
      itemCount: daftarItem.length,
      itemBuilder: (context, i) {
        final item = daftarItem[i];
        return ListTile(
          leading: Hero(
            tag: 'ikon-${item.nama}',
            child: CircleAvatar(
              backgroundColor: item.warna,
              child: Icon(item.ikon, color: Colors.white),
            ),
          ),
          title: Text(item.nama),
          trailing: const Icon(Icons.chevron_right),
          onTap: () {
            Navigator.pushNamed(context, '/detail', arguments: item);
          },
        );
      },
    );
  }
}

class DetailPage extends StatelessWidget {
  final Item item;

  const DetailPage({super.key, required this.item});

  @override
  Widget build(BuildContext context) {
    final tema = Theme.of(context);
    return Scaffold(
      appBar: AppBar(title: Text(item.nama)),
      body: Center(
        child: Column(
          mainAxisSize: MainAxisSize.min,
          children: [
            Hero(
              tag: 'ikon-${item.nama}',
              child: CircleAvatar(
                radius: 64,
                backgroundColor: item.warna,
                child: Icon(item.ikon, size: 64, color: Colors.white),
              ),
            ),
            const SizedBox(height: 16),
            Text(item.nama, style: tema.textTheme.headlineMedium),
            Text('Detail untuk ${item.nama}', style:
tema.textTheme.bodyMedium),
          ],
        ),
      ),
    );
  }
```

```dart
}

class AnimasiTab extends StatefulWidget {
 const AnimasiTab({super.key});

 @override
 State<AnimasiTab> createState() => _AnimasiTabState();
}

class _AnimasiTabState extends State<AnimasiTab>
   with SingleTickerProviderStateMixin {
 bool _besar = false;
 int _hitung = 0;
 bool _tampilKartu = true; // Latihan 5: state Opacity
 late final AnimationController _putar;

 @override
 void initState() {
   super.initState();
   _putar = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 2),
   )..repeat();
 }

 @override
 void dispose() {
   _putar.dispose();
   super.dispose();
 }

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);
   final warna = tema.colorScheme;

   return ListView(
     padding: const EdgeInsets.all(16),
     children: [
       Text('1. AnimatedContainer', style: tema.textTheme.titleMedium),
       const SizedBox(height: 8),
       Center(
         child: AnimatedContainer(
           duration: const Duration(milliseconds: 500),
           curve: Curves.easeInOut,
           width: _besar ? 200 : 100,
           height: 100,
           decoration: BoxDecoration(
             color: _besar ? warna.primary : warna.tertiary,
             borderRadius: BorderRadius.circular(_besar ? 50 : 8),
           ),
         ),
       ),
```

```
      TextButton(
        onPressed: () => setState(() => _besar = !_besar),
        child: const Text('Ubah bentuk'),
      ),
      const Divider(),
      Text('2. AnimatedSwitcher', style: tema.textTheme.titleMedium),
      Center(
        child: AnimatedSwitcher(
          duration: const Duration(milliseconds: 300),
          transitionBuilder: (child, anim) {
            return ScaleTransition(scale: anim, child: child);
          },
          child: Text(
            '$_hitung',
            key: ValueKey(_hitung),
            style: tema.textTheme.displayMedium,
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _hitung++),
        child: const Text('Tambah'),
      ),
      const Divider(),
      Text('3. AnimationController', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: RotationTransition(
          turns: _putar,
          child: Icon(Icons.settings, size: 64, color: warna.primary),
        ),
      ),
      TextButton(
        onPressed: () {
          if (_putar.isAnimating) {
            _putar.stop();
          } else {
            _putar.repeat();
          }
          setState(() {});
        },
        child: Text(_putar.isAnimating ? 'Berhenti' : 'Putar'),
      ),
      const Divider(),
      // Latihan 5: AnimatedOpacity
      Text('4. AnimatedOpacity', style: tema.textTheme.titleMedium),
      const SizedBox(height: 8),
      Center(
        child: AnimatedOpacity(
          opacity: _tampilKartu ? 1.0 : 0.0,
          duration: const Duration(milliseconds: 500),
          child: Card(
            color: warna.secondaryContainer,
```

```
            child: const Padding(
              padding: EdgeInsets.all(16.0),
              child: Text('Kartu dengan Efek Opacity'),
            ),
          ),
        ),
      ),
      TextButton(
        onPressed: () => setState(() => _tampilKartu = !_tampilKartu),
        child: Text(_tampilKartu ? 'Sembunyikan' : 'Tampilkan'),
      ),
    ],
  );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

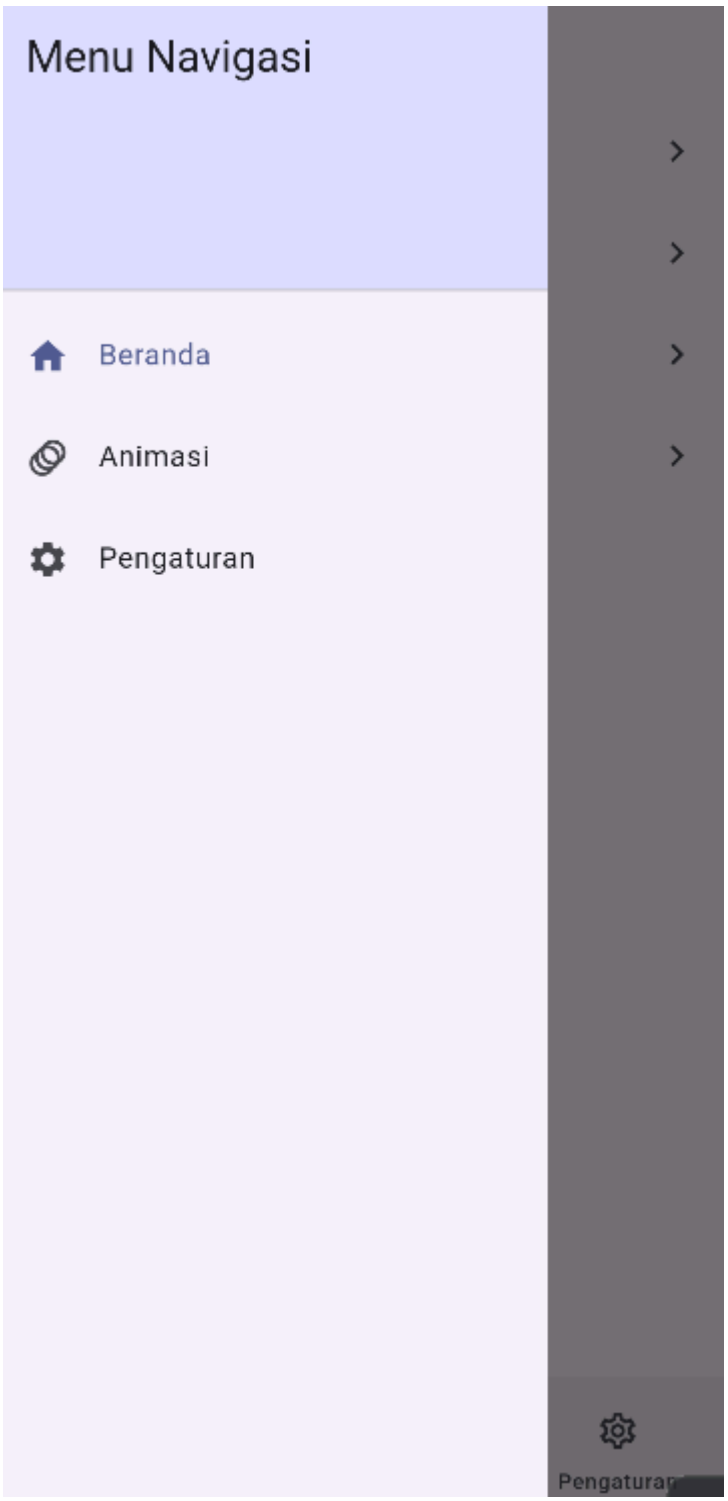
 // Latihan 1: Simpan tema ke SharedPreferences
 Future<void> _simpanTema(bool isDark) async {
   final prefs = await SharedPreferences.getInstance();
   await prefs.setBool('isDark', isDark);
 }

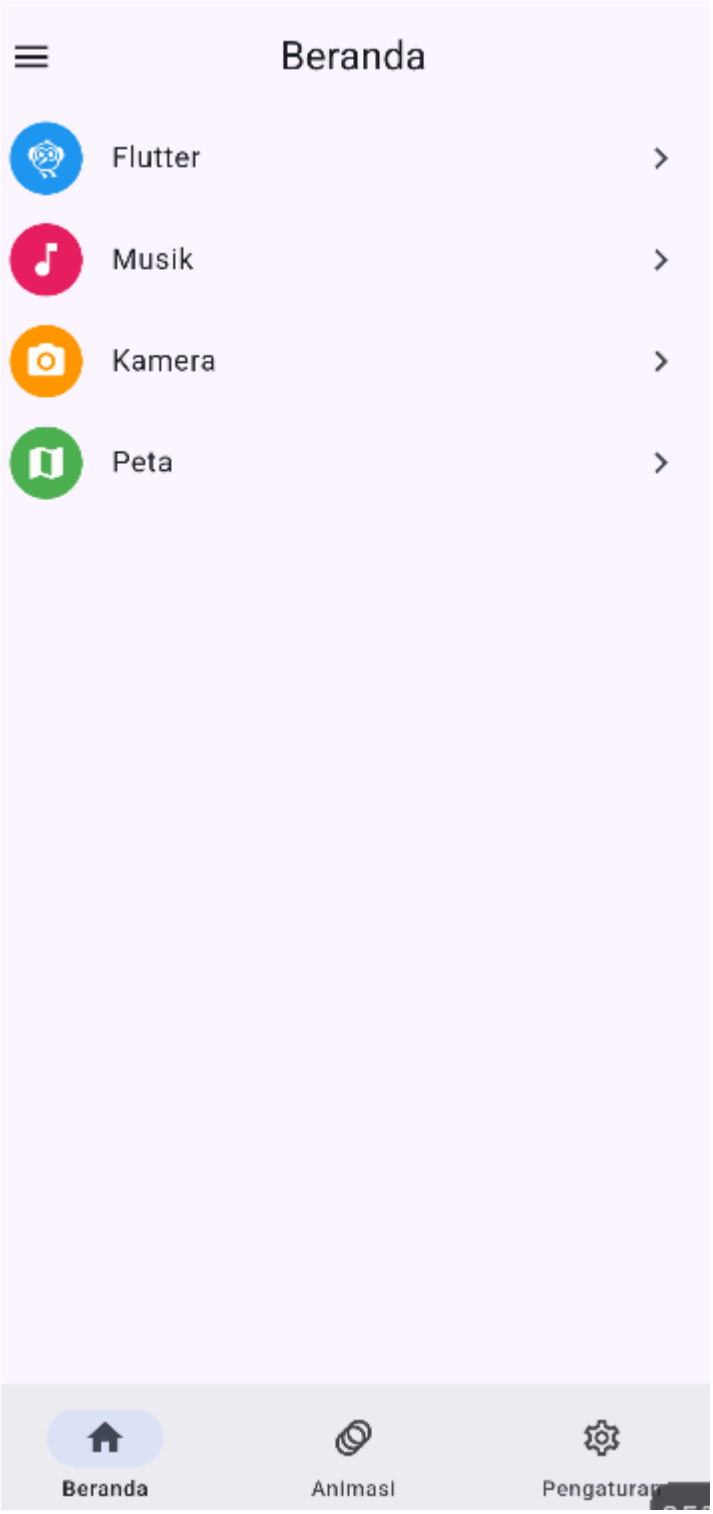
 Future<void> _simpanWarna(Color color) async {
   final prefs = await SharedPreferences.getInstance();
   await prefs.setInt('seedColor', color.value);
 }

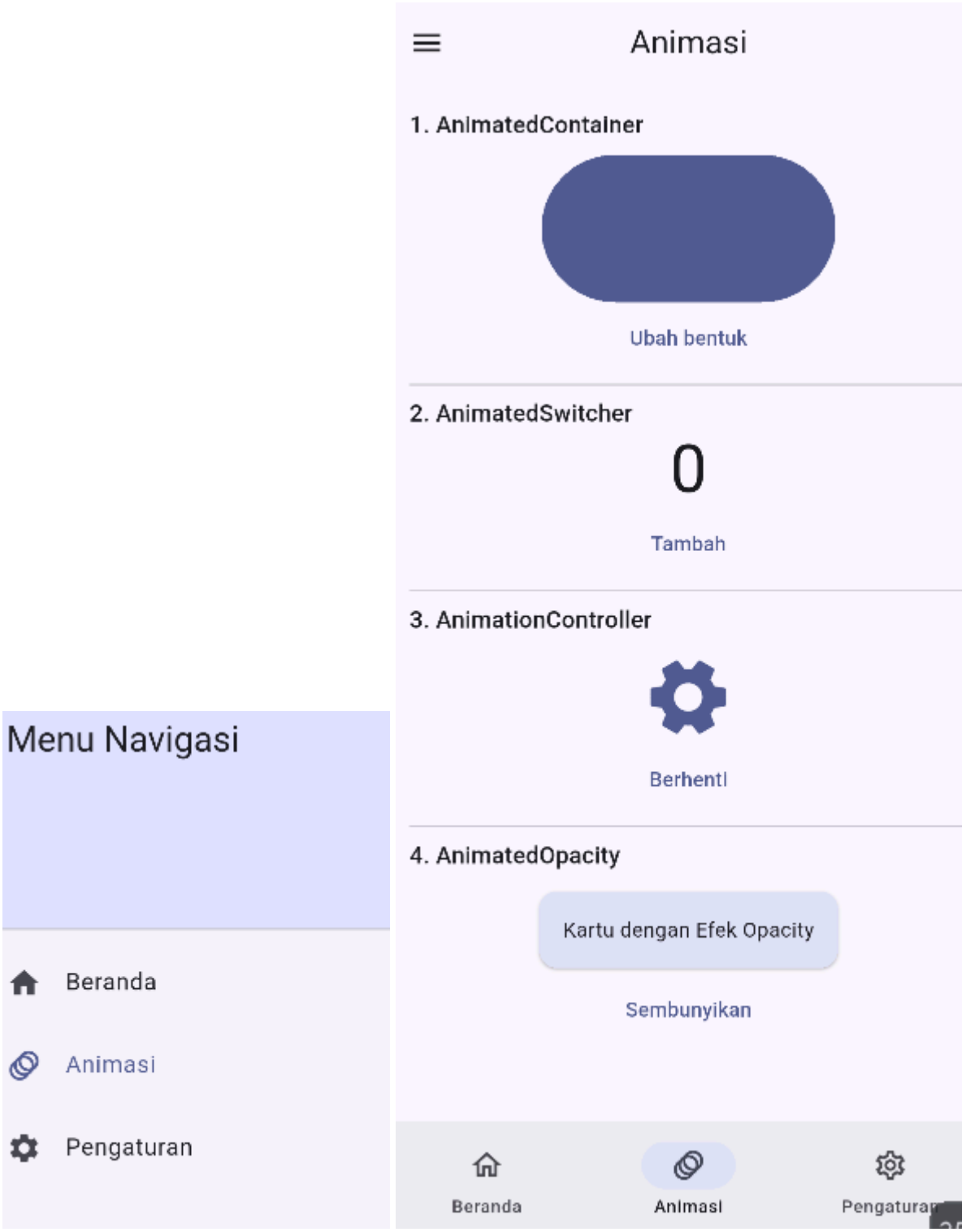
 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
               _simpanTema(v);
             },
           ),
           const SizedBox(height: 8),
           Text(
             'Warna tema',
             style: Theme.of(context).textTheme.titleMedium,
           ),
           const SizedBox(height: 8),
           Wrap(
```

```
            spacing: 12,
            children: [
              for (final c in pilihanWarna)
                GestureDetector(
                  onTap: () {
                    seedColor.value = c;
                    _simpanWarna(c);
                  },
                  child: CircleAvatar(
                    backgroundColor: c,
                    child: seedColor.value == c
                        ? const Icon(Icons.check, color: Colors.white)
                        : null,
                  ),
                ),
            ],
          ),
        ],
      );
    },
  );
 }
}
```

Hasil :

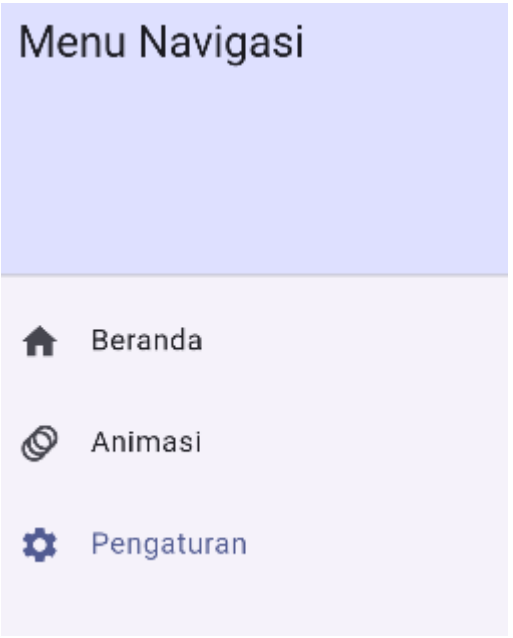

Menu Navigasi
Beranda
Animasi
Pengaturan
Pengaturan


Beranda
Flutter
Musik
Kamera
Peta
Beranda
Animasi
Pengaturan


Menu Navigasi
Beranda
Animasi
Pengaturan
Animasi
1. AnimatedContainer
Ubah bentuk
2. AnimatedSwitcher
0
Tambah
3. AnimationController
Berhenti
4. AnimatedOpacity
Kartu dengan Efek Opacity
Sembunyikan
Beranda
Animasi
Pengaturan

4. AnimatedOpacity

Tampilkan


Menu Navigasi
Beranda
Animasi
Pengaturan


Pengaturan
Mode gelap
Warna tema
Beranda
Animasi
Pengaturan

Tugas :

```dart
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

final themeMode = ValueNotifier<ThemeMode>(ThemeMode.light);
final seedColor = ValueNotifier<Color>(Colors.teal);

const pilihanWarna = [
 Colors.teal,
 Colors.blue,
 Colors.deepOrange,
 Colors.purple,
];

ThemeData buatTema(Color seed, Brightness brightness) {
 return ThemeData(
   useMaterial3: true,
   colorSchemeSeed: seed,
   brightness: brightness,
   appBarTheme: const AppBarTheme(centerTitle: true),
 );
}

class Destinasi {
 final String nama;
 final String lokasi;
 final String deskripsi;
 final IconData ikon;

 const Destinasi(this.nama, this.lokasi, this.deskripsi, this.ikon);
}

const daftarWisata = [
 Destinasi('Candi Borobudur', 'Magelang', 'Candi Buddha terbesar di
dunia.', Icons.account_balance),
 Destinasi('Pantai Kuta', 'Bali', 'Pantai pasir putih dengan pemandangan
matahari terbenam.', Icons.surfing),
 Destinasi('Gunung Bromo', 'Jawa Timur', 'Gunung berapi aktif dengan
pemandangan matahari terbit.', Icons.landscape),
 Destinasi('Danau Toba', 'Sumatera Utara', 'Danau vulkanik terbesar di
Indonesia.', Icons.water),
];

void main() async {
 WidgetsFlutterBinding.ensureInitialized();
 final prefs = await SharedPreferences.getInstance();

 final isDark = prefs.getBool('isDark') ?? false;
 themeMode.value = isDark ? ThemeMode.dark : ThemeMode.light;

 final colorValue = prefs.getInt('seedColor');
 if (colorValue != null) {
   seedColor.value = Color(colorValue);
```

```dart
 }

 runApp(const KatalogApp());
}

class KatalogApp extends StatelessWidget {
 const KatalogApp({super.key});

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return MaterialApp(
         title: 'Katalog Wisata Interaktif',
         debugShowCheckedModeBanner: false,
         theme: buatTema(seedColor.value, Brightness.light),
         darkTheme: buatTema(seedColor.value, Brightness.dark),
         themeMode: themeMode.value,
         initialRoute: '/',
         routes: {
           '/': (context) => const UtamaShell(),
         },
         onGenerateRoute: (settings) {
           if (settings.name == '/detail') {
             final destinasi = settings.arguments as Destinasi;
             return MaterialPageRoute(
               builder: (context) => DetailWisataPage(destinasi:
destinasi),
               settings: settings,
             );
           }
           return null;
         },
       );
     },
   );
 }
}

class UtamaShell extends StatefulWidget {
 const UtamaShell({super.key});

 @override
 State<UtamaShell> createState() => _UtamaShellState();
}

class _UtamaShellState extends State<UtamaShell> {
 int _index = 0;

 static const _halaman = [
   KatalogTab(),
   PromoAnimasiTab(),
```

```
    PengaturanTab(),
  ];

  static const _judul = ['Katalog Wisata', 'Promo & Animasi', 'Pengaturan'];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text(_judul[_index])),
      body: IndexedStack(index: _index, children: _halaman),
      bottomNavigationBar: NavigationBar(
        selectedIndex: _index,
        onDestinationSelected: (i) => setState(() => _index = i),
        destinations: const [
          NavigationDestination(
            icon: Icon(Icons.explore_outlined),
            selectedIcon: Icon(Icons.explore),
            label: 'Wisata',
          ),
          NavigationDestination(
            icon: Icon(Icons.local_offer_outlined),
            selectedIcon: Icon(Icons.local_offer),
            label: 'Promo',
          ),
          NavigationDestination(
            icon: Icon(Icons.settings_outlined),
            selectedIcon: Icon(Icons.settings),
            label: 'Pengaturan',
          ),
        ],
      ),
    );
  }
}

class KatalogTab extends StatelessWidget {
  const KatalogTab({super.key});

  @override
  Widget build(BuildContext context) {
    return ListView.builder(
      padding: const EdgeInsets.all(8),
      itemCount: daftarWisata.length,
      itemBuilder: (context, i) {
        final item = daftarWisata[i];
        final colorScheme = Theme.of(context).colorScheme;

        return Card(
          margin: const EdgeInsets.symmetric(vertical: 6, horizontal: 8),
          child: ListTile(
            leading: Hero(
              tag: 'wisata-${item.nama}',
              child: CircleAvatar(
```

```
              backgroundColor: colorScheme.primaryContainer,
              child: Icon(item.ikon, color:
colorScheme.onPrimaryContainer),
            ),
          ),
          title: Text(item.nama, style:
Theme.of(context).textTheme.titleMedium),
          subtitle: Text(item.lokasi, style:
Theme.of(context).textTheme.bodyMedium),
          trailing: const Icon(Icons.arrow_forward_ios, size: 16),
          onTap: () {
            Navigator.pushNamed(context, '/detail', arguments: item);
          },
        ),
      );
    },
  );
 }
}

class DetailWisataPage extends StatelessWidget {
 final Destinasi destinasi;

 const DetailWisataPage({super.key, required this.destinasi});

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);

   return Scaffold(
     appBar: AppBar(title: Text(destinasi.nama)),
     body: Padding(
       padding: const EdgeInsets.all(24.0),
       child: Column(
         children: [
           Center(
             child: Hero(
               tag: 'wisata-${destinasi.nama}',
               child: CircleAvatar(
                 radius: 70,
                 backgroundColor: tema.colorScheme.primaryContainer,
                 child: Icon(destinasi.ikon, size: 70, color:
tema.colorScheme.onPrimaryContainer),
               ),
             ),
           ),
           const SizedBox(height: 24),
           Text(destinasi.nama, style: tema.textTheme.headlineSmall),
           const SizedBox(height: 8),
           Chip(
             avatar: const Icon(Icons.location_on, size: 16),
             label: Text(destinasi.lokasi),
           ),
```

```
            const SizedBox(height: 16),
            Text(
              destinasi.deskripsi,
              style: tema.textTheme.bodyLarge,
              textAlign: TextAlign.center,
            ),
          ],
        ),
      ),
    );
  }
}

class PromoAnimasiTab extends StatefulWidget {
 const PromoAnimasiTab({super.key});

 @override
 State<PromoAnimasiTab> createState() => _PromoAnimasiTabState();
}

class _PromoAnimasiTabState extends State<PromoAnimasiTab>
    with SingleTickerProviderStateMixin {
 bool _expandedPromo = false;
 late final AnimationController _pulseController;

 @override
 void initState() {
   super.initState();
   // Animasi Eksplisit
   _pulseController = AnimationController(
     vsync: this,
     duration: const Duration(seconds: 1),
     lowerBound: 0.9,
     upperBound: 1.1,
   )..repeat(reverse: true);
 }

 @override
 void dispose() {
   _pulseController.dispose();
   super.dispose();
 }

 @override
 Widget build(BuildContext context) {
   final tema = Theme.of(context);

   return ListView(
     padding: const EdgeInsets.all(16),
     children: [
       Text('Animasi Eksplisit (Banner Diskon Pulsating)', style:
tema.textTheme.titleMedium),
       const SizedBox(height: 12),
```

```
      ScaleTransition(
        scale: _pulseController,
        child: Card(
          color: tema.colorScheme.tertiaryContainer,
          child: Padding(
            padding: const EdgeInsets.all(20),
            child: Column(
              children: [
                Icon(Icons.local_activity, size: 48, color:
tema.colorScheme.onTertiaryContainer),
                const SizedBox(height: 8),
                Text(
                  'PROMO LIBURAN 50%',
                  style: tema.textTheme.headlineSmall?.copyWith(
                    color: tema.colorScheme.onTertiaryContainer,
                  ),
                ),
              ],
            ),
          ),
        ),
      ),
      const SizedBox(height: 24),
      const Divider(),
      Text('Animasi Implisit (Kupon Diskon)', style:
tema.textTheme.titleMedium),
      const SizedBox(height: 12),
      Center(
        child: AnimatedContainer(
          duration: const Duration(milliseconds: 400),
          curve: Curves.easeInOut,
          width: _expandedPromo ? 280 : 180,
          padding: const EdgeInsets.all(16),
          decoration: BoxDecoration(
            color: _expandedPromo ? tema.colorScheme.primary :
tema.colorScheme.surfaceContainerHighest,
            borderRadius: BorderRadius.circular(_expandedPromo ? 24 : 8),
          ),
          child: Column(
            children: [
              Text(
                _expandedPromo ? 'KUPON AKTIF: WISH50' : 'Klik untuk
klaim',
                style: TextStyle(
                  color: _expandedPromo ? tema.colorScheme.onPrimary :
tema.colorScheme.onSurfaceVariant,
                  fontWeight: FontWeight.bold,
                ),
              ),
            ],
          ),
        ),
      ),
```

```
      const SizedBox(height: 8),
      Center(
        child: ElevatedButton(
          onPressed: () => setState(() => _expandedPromo =
!_expandedPromo),
          child: Text(_expandedPromo ? 'Tutup Kupon' : 'Klaim Kupon'),
        ),
      ),
    ],
  );
 }
}

class PengaturanTab extends StatelessWidget {
 const PengaturanTab({super.key});

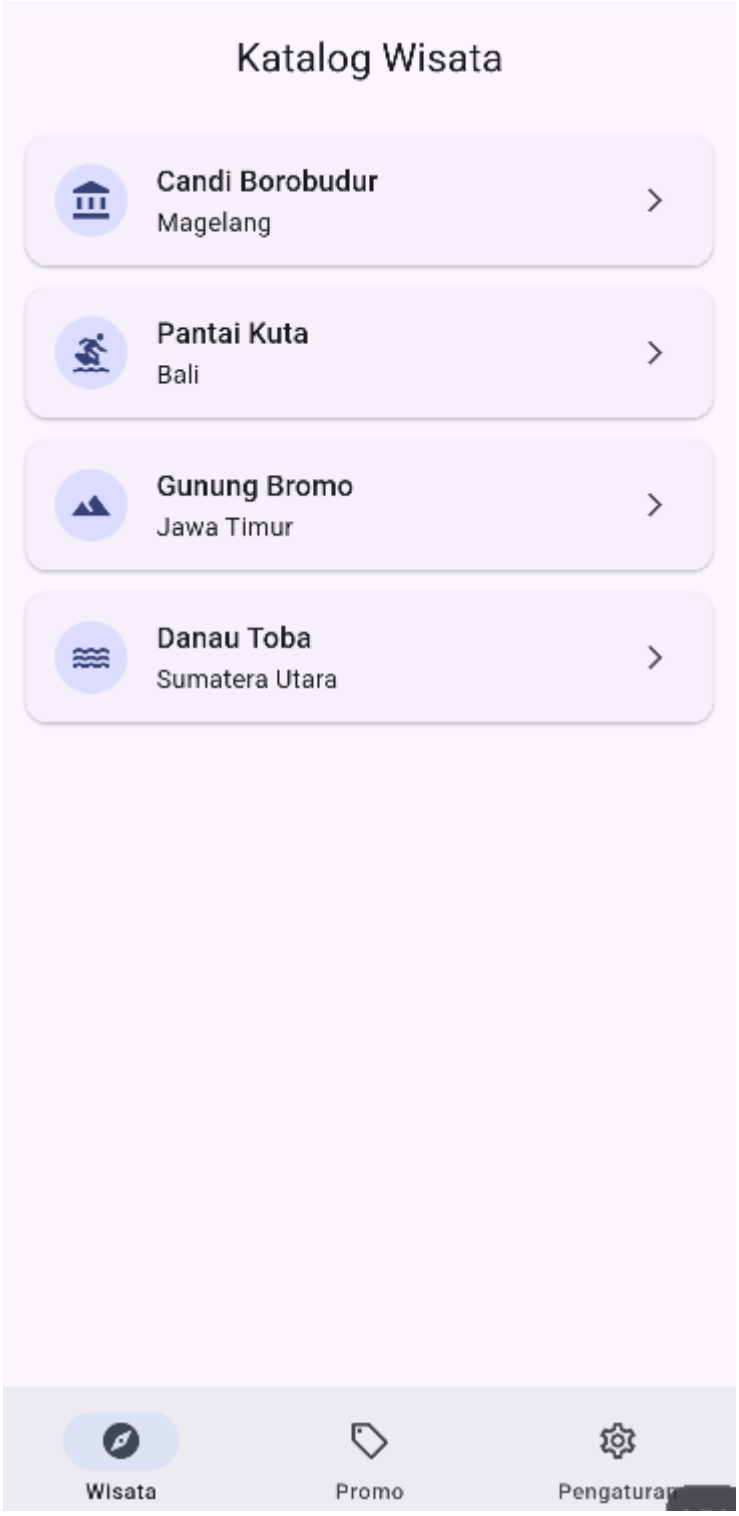
 Future<void> _simpanTema(bool isDark) async {
   final prefs = await SharedPreferences.getInstance();
   await prefs.setBool('isDark', isDark);
 }

 Future<void> _simpanWarna(Color color) async {
   final prefs = await SharedPreferences.getInstance();
   await prefs.setInt('seedColor', color.value);
 }

 @override
 Widget build(BuildContext context) {
   return ListenableBuilder(
     listenable: Listenable.merge([themeMode, seedColor]),
     builder: (context, _) {
       return ListView(
         padding: const EdgeInsets.all(16),
         children: [
           SwitchListTile(
             title: const Text('Mode Gelap'),
             value: themeMode.value == ThemeMode.dark,
             onChanged: (v) {
               themeMode.value = v ? ThemeMode.dark : ThemeMode.light;
               _simpanTema(v);
             },
           ),
           const SizedBox(height: 12),
           Text('Warna Tema Utama', style:
Theme.of(context).textTheme.titleMedium),
           const SizedBox(height: 12),
           Wrap(
             spacing: 12,
             children: [
               for (final c in pilihanWarna)
                 GestureDetector(
                   onTap: () {
                     seedColor.value = c;
```

```
                    _simpanWarna(c);
                  },
                  child: CircleAvatar(
                    backgroundColor: c,
                    child: seedColor.value == c
                        ? const Icon(Icons.check, color: Colors.white)
                        : null,
                  ),
                ),
              ],
            ),
          ],
        );
      },
    );
  }
}
```

Hasil :


Katalog Wisata
Candi Borobudur
Magelang
Pantai Kuta
Bali
Gunung Bromo
Jawa Timur
Danau Toba
Sumatera Utara
Wisata
Promo
Pengaturan

Candi Borobudur



Candi Borobudur

Magelang

Candi Buddha terbesar di dunia.

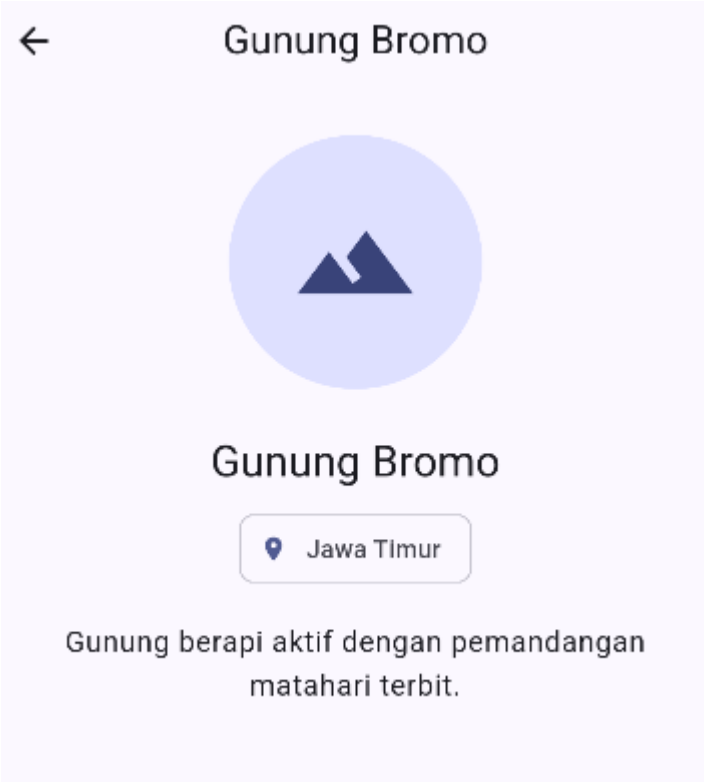
Pantai Kuta

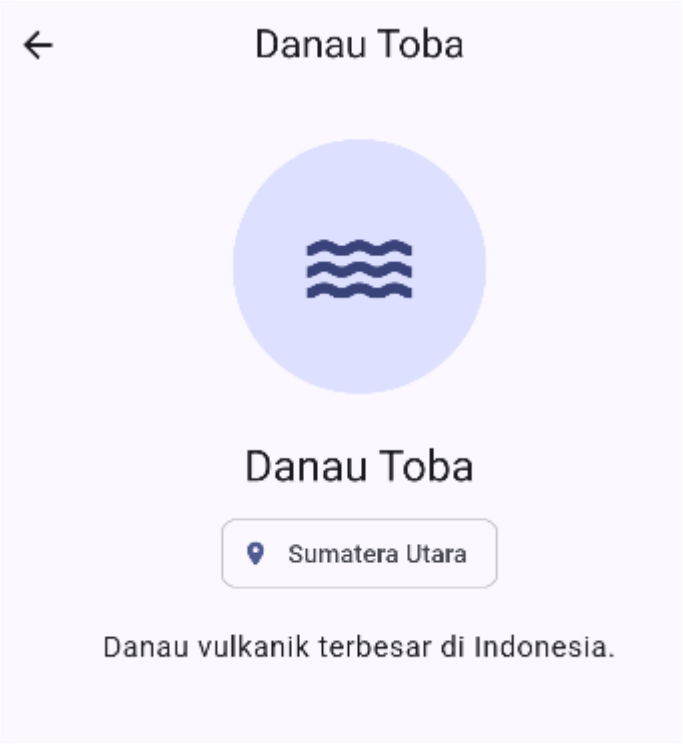


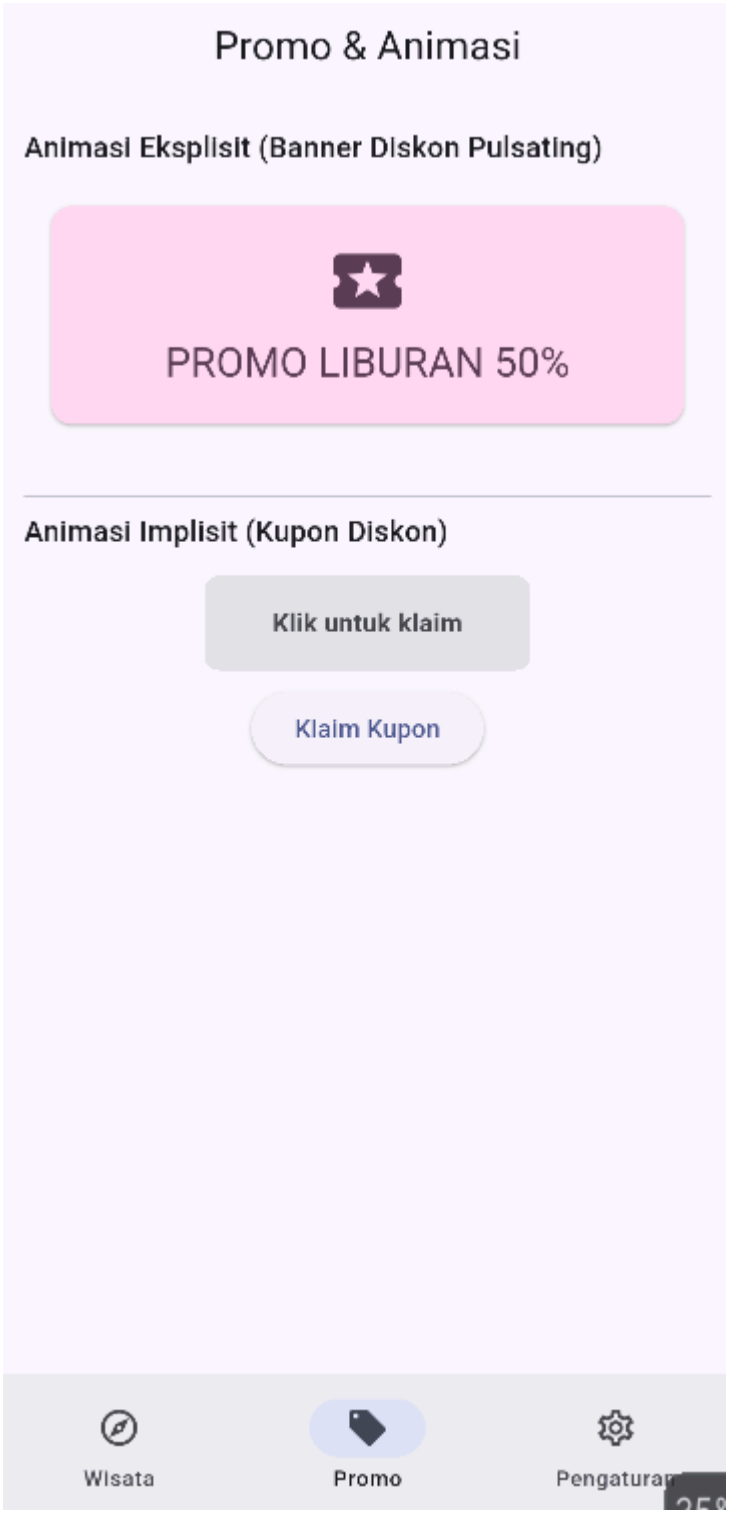
Pantai Kuta

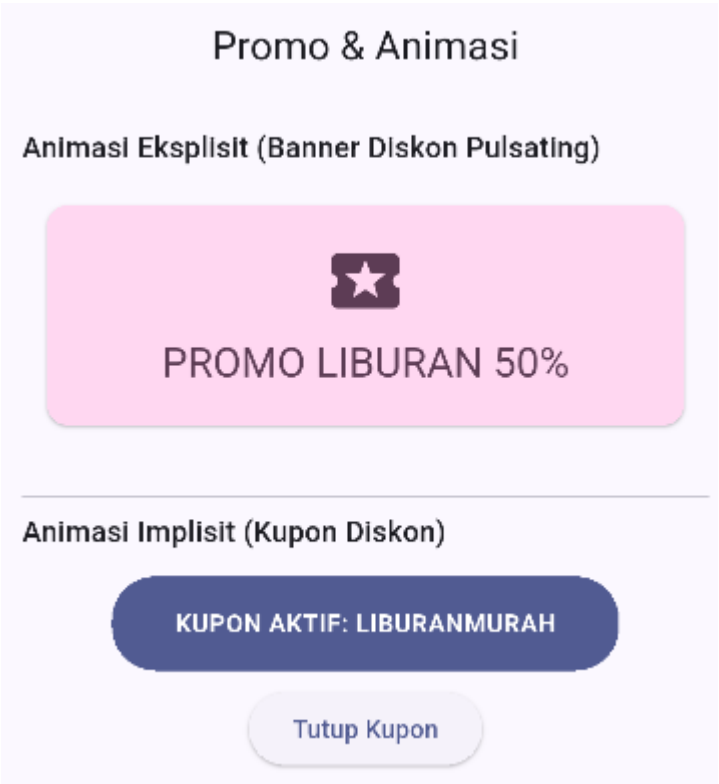
Bali

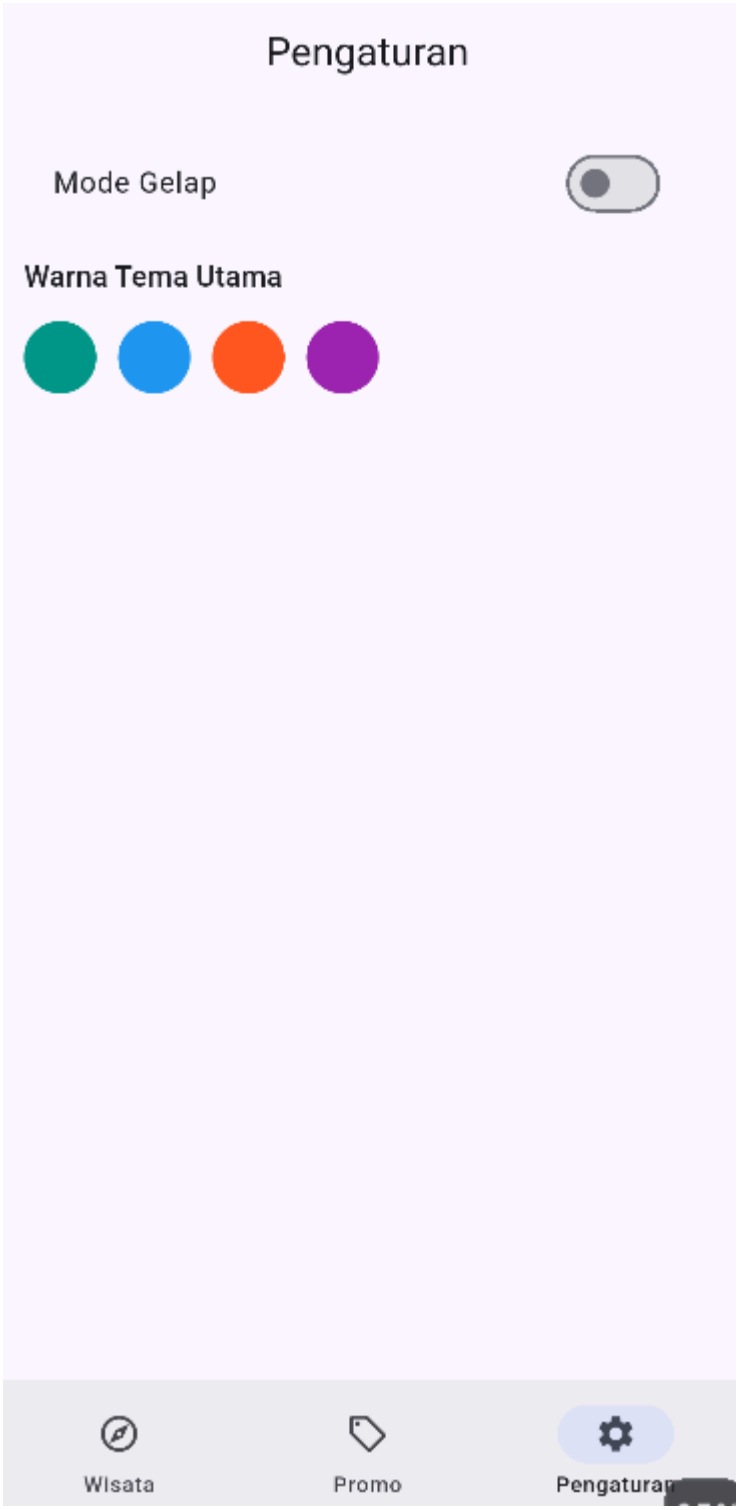
Pantai pasir putih dengan pemandangan matahari terbenam.


Gunung Bromo
Gunung Bromo
Jawa TImur
Gunung berapi aktif dengan pemandangan matahari terbit.


Danau Toba
Danau Toba
Sumatera Utara
Danau vulkanik terbesar di Indonesia.


Promo & Animasi
Animasi Eksplisit (Banner Diskon Pulsating)
PROMO LIBURAN 50%
Animasi Implisit (Kupon Diskon)
Klik untuk klaim
Klaim Kupon
Wisata
Promo
Pengaturan


Promo & Animasi
Animasi Eksplisit (Banner Diskon Pulsating)
PROMO LIBURAN 50%
Animasi Implisit (Kupon Diskon)
KUPON AKTIF: LIBURANMURAH
Tutup Kupon


Pengaturan
Mode Gelap
Warna Tema Utama
Wisata
Promo
Pengaturan